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Campfire display with augmented reality display

Abstract

A method of using a system for generating a centrally located floating image display includes displaying a first image with a first display, receiving the first image with a first reflector, reflecting the first image with the first reflector, displaying, a second image with a second display, receiving the second image with a second reflector, reflecting the second image with the second reflector, displaying first private information to the first passenger and second private information to the second passenger with a transparent display positioned between the first passenger and the first reflector and between the second passenger and the second reflector, receiving input from the first and second passengers with the system controller, collecting images with an external scene camera, and displaying general information with an augmented reality display.

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Background/Summary

INTRODUCTION

(1) The present disclosure relates to a system for generating a floating image viewable by a plurality of passengers within a vehicle.

(2) Current entertainment systems within vehicles generally comprise a screen or monitor that is mounted within the vehicle for viewing by the passengers. Some systems include smaller individual screens, wherein each passenger has a screen for their personal viewing. Current systems that provide virtual holographic images do not include the ability for annotation and for information that cannot be embedded within the virtual holographic image to be presented with the

virtual holographic image. In addition, current systems do not include tactile properties that allow a passenger to interact with the virtual holographic image, such as by making selections or choosing different images to view. Known systems incorporate inverse head-up-display architectures that use beams splitters that must be attached to structure within the vehicle compartment and must be constantly re-adjusted to accommodate height and position variations of the passengers within the vehicle compartment.

(3) While current systems achieve their intended purpose, there is a need for a new and improved system for providing a floating three-dimensional image that appears centrally located within the vehicle to all the passengers within the vehicle.

SUMMARY

(4) According to several aspects of the present disclosure, a method of using a system for generating a centrally located floating three-dimensional image display for a plurality of passengers positioned within a vehicle compartment, includes displaying, with a first display of an image chamber in communication with a system controller, a first image, receiving, with a first reflector individually associated with a first passenger, the first image from the first display, reflecting, with the first reflector, the first image to the first passenger, wherein the first passenger perceives the first image floating at a central location within the image chamber, displaying, with a second display of the image chamber in communication with a system controller, a second image, receiving, with a second reflector individually associated with a second passenger, the second image from the second display, reflecting, with the second reflector, the second image to the second passenger, wherein the second passenger perceives the second image floating at the central location within the image chamber, displaying, with a transparent display in communication with the system controller and positioned between eyes of the first passenger and the first reflector and between the eyes of the second passenger and the second reflector, first private information to the first passenger within an image plane positioned in front of the first image floating at the central location within the image chamber and second private information to the second passenger within an image plane positioned in front of the second image floating at the central location within the image chamber, receiving, with the system controller, input from the first passenger and the second passenger, collecting, with an external scene camera, images of an external environment outside the vehicle compartment, and displaying, with an augmented reality display in communication with the system controller and positioned within the vehicle compartment remotely from the image chamber, general information related to at least one of the first image, the second image, the first and second private information displayed on the transparent display and the input received from the first passenger and the second passenger.

(5) According to another aspect, the augmented reality display includes a transparent substrate, having light emitting particles dispersed therein, positioned on a window within the vehicle compartment, the displaying, with the augmented reality display positioned within the vehicle compartment remotely from the image chamber, general information related to the first image, the private information displayed on the transparent display and the input received from the first passenger including, generating, with a primary graphic projection device in communication with the system controller, a first set of images upon the window within the vehicle compartment based on visible light, wherein the first set of images are displayed upon a primary area of the window, generating, with a secondary graphic projection device in communication with the system controller, a second set of images upon a secondary area of the window based on an excitation light, wherein the light emitting particles in the windscreen emit visible light in response to absorbing the excitation light, and wherein the first set of images displayed upon the primary area of the windscreen cooperate with the second set of images displayed upon the secondary area of the windscreen to create an edge-to-edge augmented reality image.

(6) According to another aspect, the receiving, with the system controller, input from the first passenger and the second passenger, further includes, receiving, with the system controller, via the

transparent display, input from the first passenger and the second passenger, receiving, with the system controller, via at least one first sensor, input comprising a position of a head and eyes of the first passenger, receiving, with the system controller, via at least one first gesture sensor, information related to gestures made by the first passenger, collecting, with the system controller, via a first microphone, audio input from the first passenger, and collecting, with the system controller, via a second microphone, audio input from the second passenger, and the method further including, broadcasting, with the system controller, via a first zonal speaker, audio output for the first passenger, and broadcasting, with the system controller, via a second zonal speaker, audio output for the second passenger.

(7) According to another aspect, the system is adapted to support interactive gaming between the first passenger and a remote opponent, the method further including, within the first image, an image of the remote opponent, including, within the first private information displayed to the first passenger on the transparent display, game information, and including, within the general information displayed on the augmented reality display, gameplay action.

(8) According to another aspect, the method further includes capturing, with an internal scene camera mounted within the vehicle compartment and in communication with the system controller, images of the augmented reality display, and sending, with the system controller, images of the augmented reality display to the remote opponent.

(9) According to another aspect, the method further includes receiving, with the system controller, via the at least one first sensor, input comprising a position of a head and eyes of the first passenger, receiving, with the system controller, via the at least one first gesture sensor, information related to gestures made by the first passenger, and sending, with the system controller, input from the first passenger collected by the transparent display, the at least one first sensor and the at least one gesture sensor to a game system for incorporation into gameplay action.

(10) According to another aspect, the method further includes supporting communication between the first passenger and the remote opponent with the first microphone and the first zonal speaker.

(11) According to another aspect, the system is adapted to support a personalized driving tour for the plurality of passengers positioned within the vehicle compartment, the method including collecting, with the system controller, information related to a first pre-determined site and a second pre-determined site located along a pre-determined path of a driving tour, collecting, with the system controller, information related to the first passenger and the second passenger, identifying, with the system controller, at least one first point of interest located between the first pre-determined site and the second pre-determined site and relevant to the first passenger and collecting information related to the at least one first point of interest, identifying, with the system controller, at least one second point of interest located between the first pre-determined site and the second pre-determined site and relevant to the second passenger and collecting information related to the at least one second point of interest, and, when stopped at the first pre-determined site and when traveling between the first pre-determined site and the second pre-determined site, and when stopped at the second pre-determined site, including, with the system controller, the information related to the at least one first point of interest within the first image and including, with the system controller, the information related to the at least one second point of interest within the second image.

(12) According to another aspect, the method further includes, when stopped at the first pre-determined site, including, with the system controller, information related to the first pre-determined site within the general information displayed by the augmented reality display.

(13) According to another aspect, the method further includes, with the system controller, receiving, via the first microphone, a command from the first passenger, receiving, via the at least one first sensor and the at least one first gesture sensor data related to gestures made by the first passenger and the direction of a gaze of the first passenger, identifying, based on the data related to gestures made by the first passenger, the direction of the gaze of the first passenger and images of

the external environment outside the vehicle compartment, an object outside of the vehicle compartment that the first passenger is looking at through the window with the augmented reality display, highlighting the object within the augmented reality display, and supporting, with the system controller, via the transparent display, interaction, by the first passenger, with the first image.

(14) According to another aspect, when the system controller determines that the object is a permanent object, the method includes collecting information, with the system controller, from remote data sources, related to the object, and including, within the first private information, textual information related to the object.

(15) According to another aspect, when the system controller determines that the object is a non-permanent object, the method includes identifying the object with object identification algorithms within the system controller, including, within the first private information, textual information related to the object.

(16) According to another aspect, the method further includes collecting, with the system controller, from remote data sources, information related to shopping interests of the first passenger, identifying an item that the first passenger may be interested in purchasing based on the information related to shopping interest of the first passenger, identifying, with the system controller, retailers that sell the item and are located one of, within a pre-determined distance, and located on a pre-determined route, collecting, with the system controller, from identified retailers, quantity and pricing information about the item, and including, within the first private information, the information related to retailers that sell the item, quantity and pricing if the item.

(17) According to another aspect, the method further includes receiving, with the system controller, via the transparent display, input from the first passenger expressing interest in the item, including, within the first image, an image of the item, and highlighting, within the augmented reality display, an identified retailer, when an identified retailer is visible through the window and the augmented reality display.

(18) According to another aspect, the method further includes receiving, via the transparent display, input from the first passenger including augmentations to the first image, receiving, with the system controller, from the first passenger, input indicating that the first passenger is finished and wants to display the first image and the augmentations within the augmented reality display, and including, within the general information displayed on the augmented reality display, the first image and augmentations by the first passenger.

(19) According to another aspect, the method further includes receiving, from the first passenger, input on a destination, including, in the second private information displayed on the transparent display for the second passenger, information about the destination, and highlighting, within the augmented reality display, the destination when the destination is visible through the window and the augmented reality display.

(20) According to several aspects of the present disclosure, a system for generating a centrally located floating three-dimensional image display for a plurality of passengers positioned within a vehicle compartment within a vehicle includes a system controller, an image chamber including a first display adapted to project a first image, a first reflector individually associated with the first display and a first one of the plurality of passengers, the first reflector adapted to receive the first image from the first display and to reflect the first image to the first passenger, wherein the first passenger perceives the first image floating at a central location within the image chamber, a second display adapted to project a second image, and a second reflector individually associated with the second display and a second one of the plurality of passengers, the second reflector adapted to receive the second image from the second display and to reflect the second image to the second passenger, wherein, the second passenger perceives the second image floating at the central location within the image chamber, and a transparent touch screen display positioned between the first reflector and the first passenger and between the second reflector and the second passenger

and adapted to display first private information to the first passenger within an image plane positioned in front of the first image floating at the central location within the image chamber and to receive input from the first passenger, and adapted to display second private information to the second passenger within an image plane positioned in front of the second image floating at the central location within the image chamber and to receive input from the second passenger, an external scene camera adapted to collect images of an external environment outside the vehicle compartment, an augmented reality display positioned within the vehicle compartment remotely from the image chamber and adapted to display general information related to at least one of the first image, first and second private information displayed on the transparent display and input received from the first and second passengers, and an internal scene camera mounted within the vehicle compartment and in communication with the system controller and adapted to capture images of the augmented reality display.

(21) According to another aspect, the augmented reality display includes a transparent substrate, having light emitting particles dispersed therein, positioned on a window within the vehicle compartment, a primary graphic projection device for generating a first set of images upon the window of the vehicle based on visible light, wherein the first set of images are displayed upon a primary area of the window, a secondary graphic projection device for generating a second set of images upon a secondary area the window of the vehicle based on an excitation light, wherein the light emitting particles in the window emit visible light in response to absorbing the excitation light, and wherein the first set of images displayed upon the primary area of the window cooperate with the second set of images displayed upon the secondary area of the window to create an edge-to-edge augmented reality view of a surrounding environment of the vehicle, a primary graphics processing unit in electronic communication with the primary graphic projection device and the system controller, and a secondary graphics processing unit in electronic communication with the secondary graphic projection device and the system controller.

(22) According to another aspect, the system is selectively moveable vertically up and down along a vertical central axis, the first display and the first reflector are unitarily and selectively rotatable about the vertical central axis, and the second display and the second reflector are unitarily and selectively rotatable about the vertical central axis, the system further including first sensors adapted to monitor a position of a head and eyes of the first passenger, wherein, the first display and first reflector are adapted to rotate in response to movement of the head and eyes of the first passenger, and second sensors adapted to monitor a position of a head and eyes of the second passenger, wherein, the second display and the second reflector are adapted to rotate in response to movement of the head and eyes of the second passenger, the system adapted to move up and down along the vertical axis in response to movement of the head and eyes of the first passenger and movement of the head and eyes of the second passenger, and a first gesture sensor adapted to gather information related to gestures made by the first passenger, and a second gesture sensor adapted to gather information related to gestures made by the second passenger, wherein, the system is adapted to receive input from the first and second passengers via data collected by the first and second gesture sensors.

(23) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

(2) FIG. **1** is a schematic side view of a system in accordance with an exemplary embodiment of the present disclosure;
(3) FIG. **2** is a schematic top view of a vehicle compartment having a system in accordance with an exemplary embodiment of the present disclosure;
(4) FIG. **3** is a schematic diagram of the system shown in FIG. **1**;
(5) FIG. **4** is a schematic top view of the system shown in FIG. **1** with a first and second passenger;
(6) FIG. **5** is a schematic perspective view of the system shown in FIG. **1**;
(7) FIG. **6** is a schematic top view of the system shown in FIG. **3**, wherein the position of the second passenger has moved;
(8) FIG. **7** is a schematic view illustrating a passenger viewing an image and annotation information through an associated beam splitter and passenger interface; and
(9) FIG. **8** is a schematic diagram of an augmented reality display in accordance with the present disclosure;
(10) FIG. **9** is an enlarged view of a portion of FIG. **8**, as indicated by the area labelled "FIG. **9**" in FIG. **8**;
(11) FIG. **10**A is a schematic view of a window having an augmented reality display thereon having a single plane display;
(12) FIG. **10**B is a schematic view of a window having an augmented reality display thereon having a dual plane display;
(13) FIG. **11** is flow chart illustrating a method in accordance with the present disclosure;
(14) FIG. **12** is schematic view of the system of the present disclosure being utilized with a gaming system;
(15) FIG. **13** is a flow chart illustrating additional steps to the method illustrated in FIG. **11** related to capturing images with an internal scene camera;
(16) FIG. **14** is a flow chart illustrating additional steps to the method illustrated in FIG. **11** wherein the system is adapted to support a personalized driving tour for the plurality of passengers positioned within the vehicle compartment;
(17) FIG. **15** is a schematic diagram of a first pre-determined stop, a second pre-determined stop and a pre-determine path for a driving tour;
(18) FIG. **16** is a flow chart illustrating additional steps to the method illustrated in FIG. **11** related to displaying information related to shopping interests of a passenger; and
(19) FIG. **17** is schematic diagram of the system wherein images related to shopping interest of a passenger are displayed.
(20) The figures are not necessarily to scale, and some features may be exaggerated or minimized, such as to show details of particular components. In some instances, well-known components, systems, materials or methods have not been described in detail in order to avoid obscuring the present disclosure. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

DETAILED DESCRIPTION

(21) The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features. As used herein, the term module refers to any hardware, software, firmware, electronic control component, processing logic, and/or processor device, individually or in any combination, including without limitation: application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described

functionality. Although the figures shown herein depict an example with certain arrangements of elements, additional intervening elements, devices, features, or components may be present in actual embodiments. It should also be understood that the figures are merely illustrative and may not be drawn to scale.

(22) As used herein, the term "vehicle" is not limited to automobiles. While the present technology is described primarily herein in connection with automobiles, the technology is not limited to automobiles. The concepts can be used in a wide variety of applications, such as in connection with aircraft, marine craft, other vehicles, and consumer electronic components.

(23) Referring to FIG. **1** and FIG. **2**, a system **10** for generating a centrally located floating three-dimensional image **12** display for a plurality of passengers **14** positioned within a vehicle, includes an image chamber **16** that includes a first display **18** in communication with a system controller **19** and is adapted to project a first three-dimensional image **12**A and a first reflector **20** individually associated with the first display **18** and a first one **14**A of the plurality of passengers **14**, and a second display **22** that is adapted to project a second three-dimensional image **12**B and a second reflector **24** individually associated with the second display **22** and a second one **14**B of the plurality of passengers **14**. As shown in FIG. **1**, the system **10** includes two displays **18**, **22**, reflectors **20**, **24** and passengers **14**A, **14**B. It should be understood that the system **10** may be adapted to accommodate any suitable number of passengers **14**.

(24) The system controller **19** is a non-generalized, electronic control device having a preprogrammed digital computer or processor, memory or non-transitory computer readable medium used to store data such as control logic, software applications, instructions, computer code, data, lookup tables, etc., and a transceiver [or input/output ports]. computer readable medium includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A "non-transitory" computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device. Computer code includes any type of program code, including source code, object code, and executable code.

(25) Referring to FIG. **2**, a vehicle compartment **26** includes a plurality of seating positions occupied by a plurality of passengers **14**A, **14**B, **14**C, **14**D. As shown, the vehicle compartment **26** includes four seating positions for four passengers **14**A, **14**B, **14**C, **14**D. Each reflector **20**, **24**, **28**, **30** is adapted to be viewed by one of the passengers **14**A, **14**B, **14**C, **14**D. Each reflector **20**, **24**, **28**, **30** is adapted to receive an image from the associated display **18**, **22**, and to reflect the image to the associated passenger **14**. The associated passenger **14** perceives the image **12** floating at a central location within the image chamber **16**. Referring again to FIG. **1**, the first reflector **20** is adapted to receive the first image **12**A from the first display **18**, as indicated by arrows **32**, and to reflect the first image **12**A to the first passenger **14**A, as indicated by arrows **34**, wherein the first passenger **14**A perceives the first image **12**A floating at a central location within the image chamber **16**, as indicated by arrows **36**. The second reflector **24** is adapted to receive the second image **12**B from the second display **22**, as indicated by arrows **38**, and to reflect the second image **12**B to the second passenger **14**B, as indicated by arrows **40**, wherein, the second passenger **14**B perceives the second image **12**B floating at the central location within the image chamber **16**, as indicated by arrows **42**.

(26) Referring to FIG. **2**, each of the four passengers **14**A, **14**B, **14**C, **14**D perceives an image **12** reflected to them by respective associated reflectors **20**, **24**, **28**, **30** and the passengers **14**A, **14**B, **14**C, **14**D perceive the image **12** reflected to them within the image chamber **16**, as indicated by lines **44**. Each of the displays **18**, **22** can project the same image to each of the reflectors **20**, **24**, **28**, **30** and thus to each of the passengers **14**A, **14**B, **14**C, **14**D. Alternatively, each of the displays **18**, **22** can display a different perspective of the same image, or a different image altogether to each of

the reflectors **20**, **24**, **28**, **30**. Thus the system **10** is capable of presenting the same floating image **12** to all the passengers **14** so they can view simultaneously, or alternatively, each passenger **14** can view a different perspective of the floating image **12** or a completely different three-dimensional image **12**.

(27) A transparent display **46** is positioned between the eyes of each of the plurality of passengers **14** and the reflectors **20**, **24**, **28**, **30**. As shown in FIG. **1**, the transparent display **46** is positioned between the first reflector **20** and the first passenger **14**A and between the second reflector **24** and the second passenger **14**B. The transparent display **46** is adapted to display information to the first and second passengers **14**A, **14**B within an image plane positioned in front of the perceived first and second images **12**A, **12**B floating at the central location within the image chamber **16**. The transparent display **46** presents first private information to the first passenger **14**A that appears within a first image plane **48**, wherein the first private information displayed on the transparent display **46** to the first passenger **14**A appears in front of the image **12**A perceived by the first passenger **14**A within the image chamber **16**. The first private information is information meant to be seen only by the first passenger **14**A. The transparent display **46** presents second private information to the second passenger **14**B that appears within a second image plane **50**, wherein second private information displayed on the transparent display **46** to the second passenger **14**B appears in front of the image **12**B perceived by the second passenger **14**B within the image chamber **16**. The second private information is information meant to be seen only by the second passenger **14**B.

(28) In an exemplary embodiment, the transparent display **46** is a transparent touch screen that is adapted to allow the plurality of passengers **14** to receive annotated information and to provide input to the system **10**. Referring to FIG. **1** and FIG. **2**, in an exemplary embodiment, the transparent display **46** includes a clear cylindrical touch screen. The clear cylindrical touch screen encircles the image chamber **16** and is thereby positioned between the eyes of the plurality of passengers **14** and the perceived image **12** floating at the central location within the image chamber **16**. In an exemplary embodiment, the transparent display **46** is an organic light-emitting diode (OLED). It should be understood, that the transparent display **46** may be other types of transparent touch screen displays known in the art.

(29) The transparent display **46** is adapted to present visible displayed information only to the passenger **14** that is directly in front of a portion of the transparent display **46**. The nature of the transparent display **46** is such that the displayed information is only displayed on a first side, the outward facing cylindrical surface, of the transparent display **46**. A second side, the inward facing cylindrical surface, of the transparent display **46** does not display information, and thus, when viewed by the other passengers **14**, allows the other passengers **14** to see through the transparent display **46**.

(30) In an exemplary embodiment, the images from each of the displays **18**, **22** are generated via holographic method, pre-computed and encoded into a hologram generator within the display **18**, **22**. In an exemplary embodiment, each display **18**, **22** is adapted to project a three-dimensional image with variable virtual image distance. Three-dimensional images with variable virtual image distance allows the system **10** to project a floating image **12** to the passengers **14** with the capability of making the floating image **12** appear closer or further away from the passengers **14**.

(31) Referring again to FIG. **1**, in an exemplary embodiment, the system is mounted to a support structure suspended from a roof **29** within the vehicle compartment **26**. Alternatively, in another exemplary embodiment, the system is mounted to a support structure, such as a pedestal, mounted to a floor **31** within the vehicle compartment **26**. In various embodiments, the system bay be retractable, wherein, when not in use, the system recesses within the roof **29** or the floor **31** within the vehicle compartment.

(32) The transparent display **46** and each of the reflectors **20**, **24**, **28**, **30** are transparent, wherein a passenger **14** can see through the transparent display **46** and an associated reflector **20**, **24**, **28**, **30**.

This allows the passenger **14** to perceive the floating image **12** at a distance beyond the reflector **20**, **24**, **28**, **30** and further, allows the passenger **14** to see through the transparent display **46** and the reflectors **20**, **24**, **28**, **30** and able to see the interior of the vehicle compartment **26** and other passengers **14** therein.

(33) In one exemplary embodiment, the transparent display **46** is an autostereoscopic display that is adapted to display stereoscopic, or three-dimensional images by adding binocular perception of three-dimensional depth without the use of special headgear, glasses, something that affects the viewer's vision, or anything for the viewer's eyes. Because headgear is not required, autostereoscopic displays are also referred to as “glasses-free 3D” or “glassesless 3D”. The autostereoscopic transparent display includes a display panel and a parallax barrier mounted to the display panel, on an outwardly facing side of the display panel facing an associated one of the plurality of passengers **14**. In an exemplary embodiment the parallax barrier that is mounted onto the transparent display **46** includes a plurality of parallel, vertical apertures, that divide the image displayed such that a left eye and a right eye of a passenger **14** viewing the autostereoscopic display see different portions of the displayed image and the passenger **14** perceives a three-dimensional image.

(34) In an exemplary embodiment, the parallax barrier that is mounted onto the transparent display **46** is selectively actuatable by a controller adapted to switch between having the parallax barrier off, wherein the parallax barrier is completely transparent, and the viewing passenger **14** sees images displayed on the transparent display **46** as two-dimensional images, and having the parallax barrier on, wherein the viewing passenger **14** sees the images displayed on the transparent display **46** as a three-dimensional images.

(35) When the parallax barrier is actuated, each of the left and right eyes of the viewing passenger **14** only see half of the displayed image, therefore, the resolution of the three-dimensional image is reduced. To improve resolution, in one exemplary embodiment, the controller is configured to implement time-multiplexing by alternately turning the parallax barrier on and off. Time-multiplexing requires the system **10** to be capable of switching the parallax barrier on and off fast enough to eliminate any perceptible image flicker by the viewing passenger **14**. Liquid crystal displays are particularly suitable for such an application.

(36) Referring to FIG. **4**, the image chamber **16** includes transparent portions **52**, **54** to allow the passengers **14** to see their associated reflector **20**, **24**, **28**, **30**. As shown, the image chamber **16** includes a first transparent portion **52** that is adapted to allow the first image **12**A reflected by the first reflector **20** to pass from the image chamber **16** outward toward the first passenger **14**A, as indicated by arrows **34** in FIG. **1**. Further, the image chamber **16** includes a second transparent portion **54** that is adapted to allow the second image **14**B reflected by the second reflector **24** to pass from the image chamber **16** outward toward the second passenger **14**B, as indicated by arrows **40** in FIG. **1**.

(37) The image chamber **16** further includes solid portions **56**, **58** that are adapted to prevent light from entering the image chamber **16** behind the first and second reflectors **20**, **24**. The image chamber **16** functions much like a Pepper's Ghost Chamber, wherein the image of an object is perceived by a viewer within a reflective surface adjacent the actual image. As discussed above, in the present disclosure, the image presented by a display **18**, **22** which is not within view of a passenger **14**, is reflected by a reflector **20**, **24**, **28**, **30** to the passenger **14**A, **14**B, **14**C, **14**D such that the passenger “sees” the image within the image chamber **16** and perceives the image **12** to be floating behind the reflective surface of the reflector **20**, **24**, **28**, **30**. If the image chamber **16** behind the reflectors **20**, **24**, **28**, **30** is exposed to ambient light, the image will not be viewable by the passengers **14**. Thus, solid portions **56**, **58** of the image chamber **16** are adapted to prevent light from entering the image chamber **16** behind the first and second reflectors **20**, **24**. Referring to FIG. **4**, the image chamber **16** includes solid overlapping panels **56**, **58** that are adapted to prevent light from entering the image chamber **16** behind the first and second reflectors **20**, **24**.

(38) Referring to FIG. **5**, in an exemplary embodiment, the system **10** is selectively moveable vertically up and down along a vertical central axis **60**, as indicated by arrow **62**. Further, each display **18**, **22** and the associated reflector **20**, **24**, **28**, **30** are unitarily and selectively rotatable about the vertical central axis **60**, as shown by arrows **64**. This allows the system **10** to adjust to varying locations of the passengers **14** within the vehicle compartment **26**.
(39) Referring to FIG. **6**, the first reflector **20** and the first display **18** are rotatable about the vertical central axis **60**, as indicated by arrow **66**. The second reflector **24** and the second display **22** are rotatable about the vertical central axis **60**, as indicated by arrow **68**. As shown in FIG. **4**, the first and second passengers **14**A, **14**B are sitting directly across from one another, and the first reflector **20** and first display **18** are positioned 180 degrees from the second reflector **24** and second display **22**. As shown in FIG. **6**, the position of the head of the second passenger **14**B has moved, and the second reflector **24** and the second display **22** have been rotated an angular distance **70** to ensure the second passenger **14**B perceives the image **12** from the second display **22** and the second reflector **24**.
(40) In an exemplary embodiment, the image chamber **16** includes first solid panels **56** positioned adjacent the first reflector **20** on either side and adapted to move unitarily with the first reflector **20** and the first display **18** as the first reflector **20** and the first display **18** rotate about the vertical central axis **60**. Second solid panels **58** are positioned adjacent the second reflector **24** on either side and are adapted to move unitarily with the second reflector **24** and the second display **22** as the second reflector **24** and the second display **22** rotate about the vertical central axis **60**. The first solid panels **56** overlap the second solid panels **58** to allow relative movement of the first solid panels **56** relative to the second solid panels **58** and to ensure that ambient light is blocked from entering the image chamber **16** behind the first and second reflectors **20**, **24** at all times.
(41) In an exemplary embodiment, each of the displays **18**, **22** and associated reflectors **20**, **24**, **28**, **30** are equipped with head tracking capability, wherein an orientation of each display **18**, **22** and associated reflector **20**, **24**, **28**, **30** changes automatically in response to movement of a head and eyes of a passenger **14** detected by a monitoring system **72**. Monitoring systems **72** within a vehicle include sensors **74** that monitor head and eye movement of a driver/passenger within the vehicle.
(42) In an exemplary embodiment, the system **10** includes at least one first sensor **74** adapted to monitor a position of a head and eyes of the first passenger **14**A. The at least one first sensor **74** may include camera and motion sensors adapted to detect the position and movement of the first passenger's head and eyes. As shown, the first sensors **74** include a camera oriented to monitor the position and movement of the head and eyes of the first passenger **14**A. The first display **18** and first reflector **20** are adapted to rotate in response to movement of the head and eyes of the first passenger **14**A. The system **10** further includes at least one second sensor **76** adapted to monitor a position of a head and eyes of the second passenger **14**B. The at least one second sensor **76** may include camera and motion sensors adapted to detect the position and movement of a passenger's head and eyes. As shown, the second sensors **76** include a camera oriented to monitor the position and movement of the head and eyes of the second passenger **14**B. The second display **22** and second reflector **24** are adapted to rotate about the vertical central axis **60** in response to movement of the head and eyes of the second passenger **14**B.
(43) Referring again to FIG. **3**, a controller **78** of the monitoring system **72** is in communication with the system controller **19** and receives information from the first sensors **74**, and in response to detection of head/eye movement by the first passenger **14**A, actuates a first motor **80** adapted to rotate the first reflector **20** and first display **18** about the vertical central axis **60**. Further, the controller **78** of the monitoring system **72** receives information from the second sensors **76**, and in response to detection of head/eye movement by the second passenger **14**B, actuates a second motor **82** adapted to rotate the second reflector **24** and second display **22** about the vertical central axis **60**.
(44) In addition to rotation of the first display **18** and first reflector **20** and the second display **22** and second reflector **24**, the system **10** is adapted to move up and down along the vertical central

axis **60** in response to movement of the head and eyes of the first passenger **14**A and movement of the head and eyes of the second passenger **14**B. The controller **78** of the monitoring system **72** receives information from the first sensors **74** and the second sensors **76**, and in response to detection of head/eye movement by the first and second passengers **14**A, **14**B, actuates a third motor **84** adapted to raise and lower the system **10** along the vertical central axis **60** to maintain optimal vertical position of the system **10** relative to the passengers **14**. Preferences may be set within the system **10** such that the system **10** maintains optimal vertical positioning relative to a designated one of the plurality of passengers **14**, or alternatively, preferences can be set such that the system **10** maintains a vertical position taking into consideration some or all of the plurality of passengers **14**.

(45) In an exemplary embodiment, the monitoring system **72** is adapted to monitor the position of a head and eyes of each one of the plurality of passengers **14**, wherein, for each of the plurality of passengers **14**, the system **10** is adapted to display information at a specific location on the transparent display **46** based on a position of the head and eyes of the passenger **14**. In another exemplary embodiment, for each of the plurality of passengers **14**, the system **10** is adapted to display information at a specific location on the transparent display **46** based on the position of the head and eyes of the passenger **14** relative to the perceived image **12** within the image chamber **16**, such that, for each of the plurality of passengers **14**, information displayed on the transparent display **46** is properly positioned relative to the perceived image **12** within the image chamber **16**.

(46) Referring to FIG. **7**, in a schematic view of a passenger **14** an associated transparent display **46** and a floating image **12**, the passenger **14** perceives the floating image **12** at a distance behind the transparent display **46**. The transparent display **46** displays information related to the floating image **12** at a proper location on the transparent display **46** so the passenger **14** sees the information at a proper location relative to the floating image **12**. As shown in FIG. **7**, the floating image **12** is of a skyline, and more specifically, of three buildings, a first building **86**, a second building **88**, and a third building **90**. The transparent display **46** displays first building information **92**, second building information **94** and third building information **96**.

(47) The first building information **92** appears in a text box and may contain information about the first building **86** as well as the option of allowing the passenger **14** to touch the first building information **92** text box to acquire additional information about the first building **86**. For example, the first building information **92** text box may contain the name of the first building **86** and the street address. The passenger **14** may opt to touch the first building information **92** text box, wherein additional information will appear on the transparent display **46**, such as the date the first building **86** was built, what type of building (office, church, arena, etc.), or statistics such as height, capacity, etc. The second building information **94** and the third building information **96** also appear in text boxes that contain similar information and the option for the passenger **14** to touch the second or third building information **94**, **96** text boxes to receive additional information about the second and third buildings **88**, **90**.

(48) The monitoring system **72** tracks the position of the passenger's **14** head **14**H and eyes **14**E and positions the first, second and third building information **92**, **94**, **96** text boxes at a location on the transparent display **46**, such that when the passenger **14** looks at the floating image **12** through the reflector **20**, **24**, **28**, **30** and the transparent display **46**, the passenger **14** sees the first, second and third building information **92**, **94**, **96** text boxes at the proper locations relative to the floating image **12**. For example, the transparent display **46** positions the first building information **92** in the passenger's line of sight, as indicated by dashed line **98**, such that the first building information **92** is perceived by the passenger **14** at a location immediately adjacent the first building **86**, as indicated at **100**. Correspondingly, the transparent display positions the second building information **94** in the passenger's line of sight, as indicated by dashed line **102**, and the third building information **96** in the passenger's line of sight, as indicated by dashed line **104**, such that the second and third building information **94**, **96** is perceived by the passenger **14** at a location

superimposed on the building, in the case of the second building **88**, as indicated at **106**, and at a location immediately adjacent the building, in the case of the third building **90**, as indicated at **108**.

(49) The monitoring system **72** continuously tracks movement of the head **14**H and eyes **14**E of the passenger **14** and adjusts the position that the first, second and third building information **92**, **94**, **96** are displayed on the transparent display **46** to ensure that the passenger **14** always perceives the first, second and third building information **92**, **94**, **96** at the proper locations **100**, **106**, **108** relative to the floating image **12**.

(50) In an exemplary embodiment, the system **10** is adapted to accept input from a passenger **14** based solely on contact between the passenger **14** and the transparent display **46**. For example, when a passenger **14** reaches out to touch a finger-tip to the transparent display **46**, the transparent display **46** takes the input based solely on the point of contact between the tip of the finger of the passenger **14** and the transparent display **46**.

(51) In another exemplary embodiment, the system **10** is adapted to accept input from a passenger **14** based on contact between the passenger **14** and the transparent display **46** and based on the location of a point of contact between the passenger **14** and the transparent display **46** relative to the perceived image **12**. For example, the monitoring system **72** tracks the movement and position of the passenger's **14** eyes **14**E and head **14**H. The transparent display **46** displays information that is perceived by the passenger **14** relative to the floating image **12**, as discussed above. When the passenger **14** touches the transparent display **46**, the passenger **14** perceives that they are touching the floating image **12**. The system **10** uses parallax compensation to correlate the actual point of contact between the finger-tip of the passenger **14** on the transparent display **46** to the location on the floating image **12** that the passenger **14** perceives they are touching.

(52) The system **10** may display, on the transparent display **46**, multiple different blocks of annotated information relative to a floating image **12**. As the passenger's **14** head **14**H and eyes **14**E move, the passenger's head **14**H and eyes **14**E will be positioned at a different distance and angle relative to the transparent display **46**, thus changing the perceived location of displayed information relative to the image **12**. By using parallax compensation techniques, such as disclosed in U.S. Pat. No. 10,318,043 to Seder, et al., hereby incorporated by reference herein, the system **10** ensures that when the passenger **14** touches the transparent display **46**, the system **10** correctly identifies the intended piece of annotated information that the passenger **14** is selecting.

(53) In another exemplary embodiment, the system **10** is adapted to accept input from a passenger **14** based on gestures made by the passenger **14** where the passenger **14** does not touch the transparent display **46**. For example, when the passenger **14** moves a hand **114**, or points to an object that is displayed on the transparent display **46** or to an object within the vehicle compartment **26** or outside of the vehicle compartment **26**.

(54) Referring again to FIG. **1**, in an exemplary embodiment, the system includes at least one first gesture sensor **110** adapted to monitor position and movement of arms, hands and fingers **114** of the first passenger **14**A and to gather data related to gestures made by the first passenger **14**A. The first gesture sensor **110** may include a camera and motion sensors adapted to detect the position and movement of the first passenger's arms, hands and fingers. As shown, the first gesture sensor **110** includes a camera oriented to monitor the position and movement of the arms, hands and fingers of the first passenger **14**A. Further, the system **10** includes a second gesture sensor **112** adapted to monitor position and movement of arms, hands and fingers of the second passenger **14**B and to gather data related to gestures made by the second passenger **14**B. The second gesture sensor **112** may include a camera and motion sensors adapted to detect the position and movement of the second passenger's arms, hands and fingers. As shown, the second gesture sensor **112** includes a camera oriented to monitor the position and movement of the arms, hands and fingers of the second passenger **14**B.

(55) The system **10** uses data collected by the first and second gesture sensors **110**, **112** to identify gestures made by the passengers **14**A, **14**B within the vehicle compartment **26**. A system controller

will use computer learning algorithms and parallax compensation techniques to interpret such gestures and identify input data, such as when a passenger **14** is pointing to an object outside the vehicle compartment **26**.

(56) In another exemplary embodiment, the system **10** is adapted to accept audio input from passengers **14** within the vehicle compartment **26**. Referring to FIG. **1**, the system **10** includes a first microphone **116**, in communication with the system controller **19**, that is adapted to collect audio input from the first passenger **14**A. Correspondingly, the system **10** includes a second microphone **118**, in communication with the system controller **19**, that is adapted to collect audio input from the second passenger **14**B. The system controller **19** is adapted to recognize pre-determined commands from the passengers **14**A, **14**B within the vehicle compartment **26** when such commands are vocalized by the passengers **14**A, **14**B and picked-up by the first and second microphones **116**, **118**.

(57) Further, the system **10** includes a first zonal speaker **120** adapted to broadcast audio output to the first passenger **14**A. The first zonal speaker **120** is adapted to broadcast audio output in a manner such that only the first passenger **14**A can hear and understand the audio output from the first zonal speaker **120**. In this manner, audio information can be broadcast, by the system controller **19**, to the first passenger **14**A that is private to the first passenger **14**A and does not disturb other passengers within the vehicle compartment **26**. The system **10** includes a second zonal speaker **122** adapted to broadcast audio output to the second passenger **14**B. The second zonal speaker **122** is adapted to broadcast audio output in a manner such that only the second passenger **14**B can hear and understand the audio output from the second zonal speaker **122**. In this manner, audio information can be broadcast, by the system controller **19**, to the second passenger **14**B that is private to the second passenger **14**B and does not disturb other passengers within the vehicle compartment **26**. The first and second zonal speakers **120**, **122** may comprise speakers that are mounted within the vehicle compartment **26** and to broadcast audio output directionally to a specified location within the vehicle compartment **26**. Further, the first and second zonal speakers **120**, **122** may comprise a wireless headset or ear-bud adapted to be worn by the passengers **14**A, **14**B.

(58) In an exemplary embodiment, the system **10** further includes an external scene camera **124** that is in communication with the system controller **19** and is adapted to capture images of an external environment outside the vehicle compartment **26**. In this manner, the system controller **19** can collect data and “see” objects, locations, destinations and points of interest immediately outside the vehicle compartment **26**.

(59) The system **10** further includes an augmented reality display **125** positioned within the vehicle compartment **26** remotely from the image chamber **16** and adapted to display general information related to at least one of the first image **12**A, first and second private information displayed on the transparent display **46** and input received from the first and second passengers **14**A, **14**B.

(60) Referring to FIG. **8** and FIG. **9**, an augmented reality display **125** includes a transparent substrate **138** affixed to a window **127** within the vehicle compartment **26** and including light emitting particles **136** embedded therein. As explained below, the augmented reality display **125** includes a primary graphic projection device **126** and a secondary graphic projection device **128** that work together to provide an augmented reality experience for the passengers **14** within the vehicle compartment **26**.

(61) The augmented reality display **125** includes one or more controllers **129** in electronic communication with the system controller **19** and the external scene camera **124**, the monitoring system **72**, a primary graphics processing unit **130** corresponding to the primary graphic projection device **126**, and a secondary graphics processing unit **132** corresponding to the secondary graphic projection device **128**. The external scene camera **124** may be cameras that obtain periodic or sequential images representing a view of a surrounding environment outside the vehicle compartment **26**. As described above, the monitoring system **72** includes one or more sensors for

determining the location of a head of a passenger **14** within the vehicle compartment **26** as well as the orientation or gaze location of the passenger's eyes.

(62) When excitation light is absorbed by the light emitting particles **136**, visible light is generated by the light emitting particles **136**. In an embodiment, the light emitting particles **136** are red, green, and blue (RGB) phosphors for full color operation, however, it is to be appreciated that monochrome or a two-color phosphor may be used as well. Referring to FIG. **10**A, the window **127** includes a primary area **140** and a secondary area **142**, where the primary graphic projection device **126** generates a first set of images **144** upon the primary area **140** of the window **127** based on visible light, and the secondary graphic projection device **128** generates a second set of images **146** upon the secondary area **142** the window **127** based on an excitation light. Specifically, the light emitting particles **136** dispersed within the transparent substrate **138** emit visible light in response to absorbing the excitation light emitted by the secondary graphic projection device **128**. The first set of images **144** displayed upon the primary area **140** of the window **127** cooperate with the second set of images **146** displayed upon the secondary area **142** of the window **127** to create an edge-to-edge augmented reality view of the environment outside the vehicle compartment **26**.

(63) The primary area **140** of the window **127** only includes a portion of the window **127** having a limited field-of-view, while the secondary area **142** of the window **127** includes a remaining portion of the window **127** that is not included as part of the primary area **140**. Combining the primary area **140** with the secondary area **142** results in an augmented reality view of the environment outside the vehicle compartment **26** that spans from opposing side edges **150** of the window **127**. The primary graphics processing unit **130** is in electronic communication with the primary graphic projection device **126**, where the primary graphics processing unit **130** translates image-based instructions from the one or more controllers **129** into a graphical representation of the first set of images **144** generated by the primary graphic projection device **126**. The first set of images **144** are augmented reality graphics **148** that are overlain and aligned with one or more objects of interest located in the environment outside the vehicle compartment **26** to provide a passenger with a virtual reality experience. In the example as shown in FIG. **10**A, the augmented reality graphics **48** include a destination pin and text that reads “0.1 mi” that are overlain and aligned with an object of interest, which is a building **152**. It is to be appreciated that the destination pin and text as shown in FIG. **10**A is merely exemplary in nature and that other types of augmented reality graphics may be used as well. Some examples of augmented reality graphics **148** include, for example, circles or boxes that surround or highlight an object of interest, arrows, warning symbols, text, numbers, colors, lines, indicators, and logos.

(64) The primary graphic projection device **126** includes a visible light source configured to generate the first set of images **144** upon the window **127**. The visible light source may be, for example, a laser or light emitting diodes (LEDs). In the embodiment as shown in FIG. **10**A, the primary area **140** of the window **127** includes a single image plane **160**, and the primary graphic projection device **126** is a single image plane augmented reality head-up display including a digital light projector (DLP) optical system

(65) In another exemplary embodiment, referring to FIG. **10**B, the primary area **140** of the window **127** includes a dual image plane **162**, and the primary graphic projection device **126** is a fixed dual image plane holographic augmented reality head-up display. In one embodiment, the dual image plane **162** may include a first, near-field image plane **162**A and a second, far-field image plane **162**B. In the present example, the first set of images **144** include both cluster content information **154** projected upon the near-field image plane **162**A and the augmented reality graphics **148** projected upon the far-field image plane **162**B. The cluster content information **154** informs the driver of the vehicle **14** of driving conditions such as, but not limited to, vehicle speed, speed limit, gear position, fuel level, current position, and navigational instructions. In the example as shown in FIG. **6**B, the cluster content information **154** includes vehicle speed.

(66) Referring again to FIG. **10**A, the secondary graphics processing unit **132** is in electronic

communication with the secondary graphic projection device **128**, where the secondary graphics processing unit **132** translates image-based instructions from the controllers **129** into a graphical representation of the second set of images **146** generated by the secondary graphic projection device **128**. The second set of images **146** include one or more primitive graphic objects **164**. In the example as shown in FIG. **10**A, the primitive graphic object **164** is a point position or pixel. Other examples of primitive graphic objects **164** include straight lines. As explained below, objects in the surrounding environment first appear in a periphery field of a driver, which is in the secondary area **142** of the window **127**, as the primitive graphic object **164**. An object of interest located in the secondary area **142** of the window **127** is first brought to the attention of the driver by highlighting the object with the secondary set of images **146** (i.e., the primitive graphic object **64**). Once the object of interest travels from the secondary area **142** of the window **127** into the primary area **140** of the window **127**, the object of interest is highlighted by the primary set of images **144** (i.e., the augmented reality graphic **148**) instead.

(67) The secondary graphic projection device **128** includes an excitation light source configured to generate the second set of images upon the window **127**. Specifically, the light emitting particles **136** dispersed within the transparent substrate **138** on the window **127** emit visible light in response to absorbing the excitation light emitted by the secondary graphic projection device **128**. In embodiments, the excitation light is either a violet light in the visible spectrum (ranging from about 380 to 450 nanometers) or ultraviolet light that induces fluorescence in the light emitting particles **136**. It is to be appreciated that since the light emitting particles **136** are dispersed throughout the transparent substrate **138** on the window **127**, there is no directionality in the fluorescence irradiated by the light emitting particles **136**. Therefore, no matter where a passenger is located within the vehicle compartment **26**, the fluorescence is always visible. In other words, no eye-box exists, and therefore the disclosed augmented reality display **125** may be used as a primary instrument. The excitation light source may be, for example, a laser or LEDs. In embodiments, the secondary graphic projection device **128** is a pico-projector having a relatively small package size and weight. A throw distance D is measured from the window **127** to a projection lens **158** of the secondary graphic projection device **128**. The throw distance D is dimensioned so that the secondary area **142** of the window **127** spans from opposing side edges **150** of the window **127** and between a top edge **166** of the window **127** to a bottom edge **168** of the window **127**.

(68) Further details of the augmented reality display are included in U.S. patent application Ser. No. 17/749,464 to Seder et al., filed on May 20, 2022 and which is hereby incorporated by reference into the present application.

(69) In an exemplary embodiment, the system **10** further includes an internal scene camera **170** mounted within the vehicle compartment **26** and in communication with the system controller **19**. The internal scene camera **170** is oriented such that the internal scene camera **170** captures images of the augmented reality display **125**.

(70) Referring to FIG. **11**, a method **200** of using a system **10** for generating a centrally located floating three-dimensional image display for a plurality of passengers **14** positioned within a vehicle compartment **26** includes, beginning at block **202**, displaying, with a first display **18** of an image chamber **16** in communication with a system controller **19**, a first image **12**A, moving to block **204**, receiving, with a first reflector **20** individually associated with a first passenger **14**A, the first image **12**A from the first display **18**, and, moving to block **206**, reflecting, with the first reflector **20**, the first image **12**A to the first passenger **14**A, wherein the first passenger **14**A perceives the first image **12**A floating at a central location within the image chamber **16**.

(71) Moving to block **208**, the method **200** further includes displaying, with a second display **22** of the image chamber **16** in communication with the system controller **19**, a second image **12**B, moving to block **210**, receiving, with a second reflector **24** individually associated with a second passenger **14**B, the second image **12**B from the second display **22**, and, moving to block **212**, reflecting, with the second reflector **24**, the second image **12**B to the second passenger **14**B,

wherein the second passenger **14**B perceives the second image **12**B floating at the central location within the image chamber **16**.

(72) Moving to block **214**, the method **200** includes displaying, with a transparent display **46** in communication with the system controller **19** and positioned between eyes of the first passenger **14**A and the first reflector **20** and between the eyes of the second passenger **14**B and the second reflector **24**, first private information to the first passenger **14**A within an image plane **48** positioned in front of the first image **12**A floating at the central location within the image chamber **16** and second private information to the second passenger **14**B within an image plane **50** positioned in front of the second image **12**B floating at the central location within the image chamber **16**.

(73) Moving to block **216**, the method **200** includes receiving, with the system controller **19**, input from the first passenger **14**A and the second passenger **14**B. Moving to block **218**, the method **200** includes collecting, with an external scene camera **124**, images of an external environment outside the vehicle compartment **26**, and, moving to block **220**, displaying, with an augmented reality display **125** in communication with the system controller **19** and positioned within the vehicle compartment **26** remotely from the image chamber **16**, general information related to at least one of the first image **12**A, the second image **12**B, the first and second private information displayed on the transparent display **46** and the input received from the first passenger **14**A and the second passenger **14**B.

(74) In an exemplary embodiment, the receiving, with the system controller **19**, input from the first passenger **14**A and the second passenger **14**B, at block **216**, further includes, moving to block **222**, receiving, with the system controller **19**, via the transparent display **46**, input from the first passenger **14**A and the second passenger **14**B, moving to block **224**, receiving, with the system controller **19**, via at least one first sensor **74**, input comprising a position of a head and eyes of the first passenger **14**A, moving to block **226**, receiving, with the system controller **19**, via at least one first gesture sensor **110**, information related to gestures made by the first passenger **14**A, moving to block **228**, collecting, with the system controller **19**, via a first microphone **116**, audio input from the first passenger **14**A, and, moving to block **230**, collecting, with the system controller **19**, via a second microphone **118**, audio input from the second passenger **14**B.

(75) Moving to block **232**, the method **200** further includes broadcasting, with the system controller **19**, via a first zonal speaker **120**, audio output for the first passenger **14**A, and, moving to block **234**, broadcasting, with the system controller **19**, via a second zonal speaker **122**, audio output for the second passenger **14**B.

(76) Referring to FIG. **12**, in an exemplary embodiment, the system is adapted to support interactive gaming between the first passenger **14**A and a remote opponent **172**. The first passenger **14**A is located within the vehicle compartment **26**, and the remote opponent **172** is located at a remote location **174**. A gaming system can be included within the system controller **19**, a game console connected to the system controller, or within a game console at the remote location **174**. The system controller communicates with the remote opponent via the cloud **176** to allow the first passenger and the remote opponent **172** to join a shared gaming environment.

(77) Wherein, the displaying the first image **12**A at block **202**, further includes, including within the first image **12**A, an image **12**RO of the remote opponent **172**. The image **12**RO of the remote opponent **172** is captured by a remote camera **176** at the remote location. Further wherein, the displaying, with the transparent display **46**, first private information at block **214** includes, including, within the first private information displayed to the first passenger **14**A on the transparent display **46**, game information. Game information displayed on the transparent display **46** may be any textual or graphic information meant only for the first passenger **14**A to see, such as personal stats, available in-game tools/weapons, etc.

(78) The displaying, with the augmented reality display **125**, general information at block **220** further includes, including, within the general information displayed on the augmented reality

display **125**, gameplay action. Thus, the first passenger **14**A sees the image **12**RO of the remote opponent **172** floating in front of the first passenger within the image chamber **16**, and receives game information relevant to the first passenger **14**A on the transparent display **46**, which may also be utilized by the first passenger **14**A to provide input to the game, such as selection of in-game options.

(79) Referring to FIG. **13**, in an exemplary embodiment, the method **200** further includes, moving to block **236**, capturing, with the internal scene camera **170** mounted within the vehicle compartment **26** and in communication with the system controller **19**, images of the augmented reality display **125**, and, moving to block **238**, sending, with the system controller **19**, images of the augmented reality display **125** to the remote opponent **172**. Thus, the first passenger **14**A, and any other passengers **14** within the vehicle compartment can watch the gameplay action on the augmented reality display **125** within the vehicle compartment **26**, and the remote opponent **172** can watch the gameplay action on a remote display **180** at the remote location **174**.

(80) In an exemplary embodiment, the receiving, with the system controller **19**, via the at least one first sensor **74**, input comprising a position of a head and eyes of the first passenger **14**A at block **224**, and, the receiving, with the system controller **19**, via the at least one first gesture sensor **110**, information related to gestures made by the first passenger **14**A, at block **226**, further includes sending, with the system controller **19**, input from the first passenger **14**A collected by the transparent display **46**, the at least one first sensor **74** and the at least one gesture sensor **110** to the game system for incorporation into gameplay action. Thus, the game system can detect when the first passenger **14**A points to objects that appear within the augmented reality display, or when the first passenger **14**A looks at a specific object within the gameplay action and incorporate such actions by the first passenger **14**A into the gameplay action. For example, when the first passenger **14**A controls a player-character within the gameplay, the actions of the player character may be controlled and triggered by actual gestures made by the first passenger **14**A.

(81) In an exemplary embodiment, the collecting, with the system controller **19**, via a first microphone **116**, audio input from the first passenger **14**A, at block **228** further includes collecting audio input from the first passenger **14**A related to gameplay, and communications from the first passenger **14**A to the remote opponent **172**. The broadcasting, with the system controller **19**, via a first zonal speaker **120**, audio output for the first passenger **14**A, at block **232** further includes broadcasting, with the system controller **19**, via the first zonal speaker **120**, audio output for the first passenger **14**A, including audible communications from the remote opponent **172** captured by a remote microphone **182** at the remote location **174**. Thus, the system supports communication between the first passenger **14**A and the remote opponent **172** with the first microphone **116**, the first zonal speaker **120**, and the remote microphone **182**.

(82) In another exemplary embodiment, the system **10** is adapted to support a personalized driving tour for the plurality of passengers **14** positioned within the vehicle compartment **26**. Referring to FIG. **14** and FIG. **15**, the method **200**, further includes, moving to block **240**, collecting, with the system controller, information related to a first pre-determined site **184** and a second pre-determined site **186** located along a pre-determined path **188** of a driving tour. The first and second pre-determined sites may be downloaded by the system controller **19** from the internet, or may be manually input by a passenger **14**, via the transparent display **46**. Once the first and second pre-determined stops **184**, **186** along the path **188** of the driving tour have been established, the method **200** includes, moving to block **242**, collecting, with the system controller **19**, information related to the first passenger **14**A and the second passenger **14**B.

(83) Information related to the first and second passengers **14**A, **14**B can be collected based on direct input by the first and second passengers **14**A, **14**B, or, such information may be pulled from a database within the system controller **19** where information related to the first and second passengers **14**A, **14**B is stored based on past behavior. Further, the system may be adapted to prompt a passenger for personal interests, wherein, when the system **10** identifies multiple potential

points of interest, the passenger **14** is allowed to select a point of interest that they are interested in. Moving to block **244**, the method **200** includes identifying, with the system controller **19**, at least one first point of interest located between the first pre-determined site **184** and the second pre-determined site **186** and relevant to the first passenger **14**A and collecting information related to the at least one first point of interest. The system controller **19**, by accessing the internet and mapping services, will look for points of interest along the pre-determined path **188** and match information related to the first passenger **14**A to points of interest to identify at least one first point of interest related to the first passenger **14**A along the path **188**, between the first and second pre-determined sites **184**, **186**.

(84) Moving to block **246**, the method **200** includes identifying, with the system controller **19**, at least one second point of interest located between the first pre-determined site **184** and the second pre-determined site **186** and relevant to the second passenger **14**B and collecting information related to the at least one second point of interest. The system controller **19**, by accessing the internet and mapping services, will look for points of interest along the pre-determined path **188** and match information related to the second passenger **14**B to points of interest to identify at least one second point of interest along the path **188**, between the first and second pre-determined sites **184**, **186**.

(85) Moving to block **248**, when the vehicle is stopped at the first pre-determined site **184**, and, when the vehicle is traveling between the first pre-determined site **184** and the second pre-determined site **186**, and when the vehicle is stopped at the second pre-determined site **186**, the method **200** further includes, including, with the system controller **19**, the information related to the at least one first point of interest within the first image **12**A and including, with the system controller **19**, the information related to the at least one second point of interest within the second image **12**B.

(86) Thus, at all times, the system controller is looking for a point of interest relevant to the passenger, and when stopped at or passing by such point of interest, displays information about that point of interest within the first and second images floating within the image chamber, and displays information about the points of interest on the transparent display **46**.

(87) Moving to block **250**, the method **200** further includes, when stopped at the first pre-determined site **184**, including, with the system controller **19**, information related to the first pre-determined site within the general information displayed by the augmented reality display **125**, and, when stopped at the second pre-determined site **186**, including, with the system controller **19**, information related to the second pre-determined site within the general information displayed by the augmented reality display **125**.

(88) Thus, when stopped at one of the pre-determined sites **184**, **186** along a pre-determined path **188**, the system **10** will display information related to the pre-determined sites **184**, **186** on the augmented reality display **125**, such that the information related to the pre-determined sites **184**, **186** is visible by all passengers **14** within the vehicle compartment **26**.

(89) For example, a pre-determined route **188** is a driving tour of a famous battlefield and the surrounding area. The first pre-determined site **184** is a wheatfield and the second pre-determined site is a battlefield, such as “Little Roundtop”. When the driving tour stops at the wheatfield (first pre-determined site **184**), the system displays information about the wheatfield on the augmented reality display for all passengers **14** within the vehicle compartment **26** to see. Simultaneously, while stopped at the wheatfield, the system **10** identifies a point of interest, such as a battlefield map, that is of interest to the first passenger **14**A, and includes the battlefield map within the first image and displays textual information about the battlefield map within the transparent display **46** for the first passenger **14**. Further, while stopped at the wheatfield, the system **10** identifies a point of interest, such as a battlefield re-enactment, that periodically takes place at the wheatfield, and is of interest to the second passenger **14**B. The system includes information about the re-enactment such as maps, pictures, etc. within the second image and displays textual information about the re-

enactment and prompts the second passenger **14**B for selections for additional information within the transparent display **46** for the second passenger **14**B.

(90) When traveling between the wheatfield (first pre-determined site **184** and the battlefield (second pre-determined site **186**), the augmented reality display is blank, but the system continues to display information for each of the passengers within the first and second images and on the transparent display. For example, while traveling between the wheatfield and the battlefield, the system **10** identifies a point of interest, such as information related to weapons and ammunition used at the time of the battle, that is of interest to the first passenger **14**A, and includes such information within the first image and displays textual information about the weapons and ammunition within the transparent display **46** for the first passenger **14**, as well as allowing the first passenger to make a selection to request additional information about a specific weapon or type of ammunition, via the touch screen feature of the transparent display **46**. Further, the system **10** identifies a point of interest, such as a site where two famous commanders argued, that is of interest to the second passenger **14**B. The system **10** includes information about the argument, such as a picture of the two commanders arguing, within the second image and displays textual information about the argument, such as why the commanders argued and who may have won the argument, within the transparent display **46** for the second passenger **14**B.

(91) Finally, when the driving tour stops at the battlefield (second pre-determined site **186**), the system displays information about the battlefield on the augmented reality display **125** for all passengers **14** within the vehicle compartment **26** to see. Simultaneously, while stopped at the battlefield **186**, the system **10** identifies a point of interest, such as conditions that soldiers were forced to endure during the battle, that is of interest to the first passenger **14**A, and includes pictures and information within the first image and displays textual information within the transparent display **46** for the first passenger **14**. Further, the system **10** identifies a point of interest, such as details of hand to hand fighting techniques used by the soldiers during the battle, that is of interest to the second passenger **14**B. The system **10** includes information about hand to hand fighting techniques within the second image and displays textual information about hand to hand fighting techniques and prompts the second passenger **14**B for selections for additional information within the transparent display **46** for the second passenger **14**B.

(92) In an exemplary embodiment, the broadcasting, with the system controller **19**, via a first zonal speaker **120**, audio output for the first passenger **14**A, at block **232** and the broadcasting, with the system controller **19**, via a second zonal speaker **122**, audio output for the second passenger **14**B, at block **234**, includes broadcasting audible information to the first passenger **14**A and the second passenger **14**B related to the pre-determined sites **184**, **186** along the path **188**, when stopped at one of the pre-determined sites **184**, **186**, and, broadcasting audible information to the first passenger **14**A related to the at least one identified first point of interest, and audible information to the second passenger **14**B related to the at least one identified second point of interest, when traveling between the pre-determined sites **184**, **186** along the path **188**.

(93) In an exemplary embodiment, the collecting, with the system controller **19**, via the first microphone **116**, audio input from the first passenger **14**A, at block **228**, includes receiving, via the first microphone **116**, a command from the first passenger that triggers the system **10** to “look” for an object which is being observed by the first passenger **14**. For example, the first passenger may say a pre-determined phrase, such as “Hey General”, which triggers the system **10**. Upon triggering of the system **10**, the receiving, with the system controller **19**, via at least one first sensor **74**, input comprising a position of a head and eyes of the first passenger **14**A at block **224** and the receiving, with the system controller **19**, via at least one first gesture sensor **110**, information related to gestures made by the first passenger **14**A at block **226**, further includes identifying, based on the data related to gestures made by the first passenger **14**A, the direction of the gaze of the first passenger **14**A and images of the external environment outside the vehicle compartment **26**, collected at block **218**, an object outside of the vehicle compartment **26** that the first passenger **14**A

is looking at through the window **127** with the augmented reality display **125**.
(94) Gaze and gesture tracking allow the system **10** to generate a vector which defines an array of GPS coordinates. The point where this vector intersects the image from the external scene camera **124** identifies the observed object and allows calculation of a distance to the observed object. Merging the captured image of the external environment with the GPS coordinate vector allows the system to pinpoint a GPS location of the observed object.
(95) Further, the displaying, with an augmented reality display **125** in communication with the system controller **19** and positioned within the vehicle compartment **26** remotely from the image chamber **16**, general information at block **220**, includes highlighting the object that is being observed by the first passenger **14**A within the augmented reality display **125**.
(96) Further, the receiving, with the system controller **19**, via the transparent display **46**, input from the first passenger **14**A at block **222** includes supporting, with the system controller **19**, via the transparent display **46**, interaction, by the first passenger **14**A, with the first image **12**A, which includes an image of the observed object, and allows the first passenger **14**A to rotate the image, or enlarge or reduce the image. Further the first passenger **14**A may via the touch screen features of the transparent display **46**, annotate the displayed object, such annotations appearing in the augmented reality display **125** for viewing by all the passengers **14** within the vehicle compartment **26**. Thus, the first passenger **14**A can see an object outside the vehicle compartment **26**, initialize the system by saying “Hey General”, wherein, the system will identify the object being observed by the first passenger **14**A, and highlight the object within the augmented reality display **125**. The system will also include the observed object within the first image **12**A, wherein the first passenger **14**A can annotate the observed object, and such annotations will appear in the augmented reality display **125** for viewing by all passengers **14** within the vehicle compartment **26**.
(97) The system controller **19** is adapted to determine if an observed object is a permanent object, such as a building or mountain, or if an observed object is a non-permanent object, such as a vehicle or a person. In an exemplary embodiment, the identifying an observed object based on the data related to gestures made by the first passenger **14**A, the direction of the gaze of the first passenger **14**A and images of the external environment outside the vehicle compartment **26**, further includes, when the system controller **19**, determines that the observed object is a permanent object, collecting information, with the system controller **19**, from remote data sources, related to the object, and including, within the first private information displayed on the transparent display **46**, textual information related to the observed object. When the system controller **19** determines that the object is a permanent object, the system controller **19** accesses the internet for information about the identified permanent object. For example, the observed permanent object may be a building, wherein the system controller **19** can access the internet and remote data bases, such as yellow pages, to obtain an address for the building, which the system controller **19** then displays on the transparent display **46**.
(98) Further, the identifying an observed object based on the data related to gestures made by the first passenger **14**A, the direction of the gaze of the first passenger **14**A and images of the external environment outside the vehicle compartment **26**, further includes, when the system controller **19**, determines that the observed object is a non-permanent object, identifying the observed object with object identification algorithms within the system controller, and including, within the first private information displayed on the transparent display **46**, textual information related to the observed object. For example, if the system controller determines that the observed object is a person, the system controller **19** displays this identification on the transparent display **46**.
(99) Referring to FIG. **16**, the method **200** further includes, moving to block **252**, collecting, with the system controller **19**, from remote data sources, information related to shopping interests of the first passenger **14**A. Referring to FIG. **17**, for example, the system controller **19** receives information collected by remote data collection databases, such as google, information about the first passenger's **14**A shopping habits via the cloud **176**. Moving to block **254**, the method **200**

includes identifying an item **190** that the first passenger **14**A may be interested in purchasing based on the information related to shopping interest of the first passenger **14**A, moving to block **256**, identifying, with the system controller **19**, retailers **192** that sell the item **190** and are located one of, within a pre-determined distance, and located on a pre-determined route, moving to block **258**, collecting, with the system controller **19**, from identified retailers **192**, quantity and pricing information about the item **190**, and, moving to block **260**, including, within the first private information displayed to the first passenger **14**A on the transparent display **46**, the information related to retailers **192** that sell the item **190**, quantity and pricing of the item **190**. As shown in the example in FIG. **17**, the first image **12**A, includes an image of an item **190**, as shown a garlic press, that may be of interest to the first passenger **14**A. The system controller **19** may display only one item, or may display multiple items to the first passenger **14**A.

(100) Moving to block **262**, the method **200** further includes receiving, with the system controller **19**, via the transparent display **46**, input from the first passenger **14**A expressing interest in the item **190**. Moving to block **264**, the method includes, including, within the first image **12**A, an image of the item **190**. The system controller **19** will display, on the transparent display **46**, information listing items that may be of interest to the first passenger **14**A, and information about such items and where they are available. When the first passenger **14**A selects, via the touch screen features of the transparent display **46**, an item **190**, here a garlic press, the system controller **19** displays an image of the garlic press within the first image **12**A. Further, moving to block **266**, the method **200** includes highlighting, within the augmented reality display **125**, an identified retailer **192**, when an identified retailer **192** is visible through the window **127** and the augmented reality display **125**. When the vehicle passes a retailer that has been identified by the system controller **19** as a retailer that sells the item **190** (garlic press), then the system controller **19** displays a highlight **194** around the retailer **192** within the augmented reality display **125**, drawing the first passenger's **14**A attention to the retailer **192**.

(101) In another exemplary embodiment, the receiving, with the system controller **19**, via the transparent display **46**, input from the first passenger **14**A at block **222**, further includes, receiving, via the transparent display **46**, input from the first passenger **14**A including augmentations to the displayed first image. For example, the first passenger **14**A may want to provide a highlight to an object within the first image, or add drawn in annotations using the touch screen features of the transparent display **46**. The system **10** may provide a series of pasteable forms that the first user **14**A can drag onto the first image **12**A.

(102) When the first passenger **14**A is finished annotating the first image **12**A, the receiving, with the system controller **19**, via the transparent display **46**, input from the first passenger **14**A at block **222**, further includes receiving, with the system controller **19**, from the first passenger **14**A, input indicating that the first passenger **14**A is finished and wants to display the first image **12**A and the annotations to the first image **12**A within the augmented reality display. Alternatively, when the first passenger **14**A is finished annotating the first image **12**A, the receiving, with the system controller **19**, via at least one first gesture sensor **110**, information related to gestures made by the first passenger **14**A at block **226**, further includes receiving, with the system controller **19**, from the first passenger **14**A, input comprising a hand gesture indicating that the first passenger **14**A is finished and wants to display the first image **12**A and the annotations to the first image **12**A within the augmented reality display **125**. The hand gesture may be a swiping motion wherein the first passenger **14**A simulates dragging and “throwing” the first image **12**A and the annotations toward the augmented reality display **125**.

(103) Finally, the displaying, with an augmented reality display **125** in communication with the system controller **19** and positioned within the vehicle compartment **26** remotely from the image chamber **16**, general information at block **220**, further includes, including, within the general information displayed on the augmented reality display **125**, the first image **12**A as augmented by the first passenger **14**A and including the annotations made to the first image **12**A by the first

passenger **14**A.

(104) In another exemplary embodiment, the receiving, with the system controller **19**, via the transparent display **46**, input from the first passenger **14**A at block **222**, further includes receiving, from the first passenger **14**A, input on a destination. For example, the destination may be the location of a friend that needs to be picked up. Further, the displaying, with the transparent display **46** in communication with the system controller **19** and positioned between eyes of the first passenger **14**A and the first reflector **20** and between the eyes of the second passenger **14**B and the second reflector **24**, second private information to the second passenger **14**B at block **214**, further includes, including, in the second private information displayed on the transparent display **46** for the second passenger **14**B, information about the destination, and the displaying, with an augmented reality display **125** in communication with the system controller **19** and positioned within the vehicle compartment **26** remotely from the image chamber **16**, general information at block **220**, further includes highlighting, within the augmented reality display **125**, the destination when the destination is visible within through the window **127** and the augmented reality display **125**. For example, when the vehicle approaches the destination where a friend is to be picked up, the system **10** may display a marker at the destination as the vehicle approaches.

(105) In another exemplary embodiment, the broadcasting, with the system controller **19**, via a first zonal speaker **120**, audio output for the first passenger **14**A at block **232**, and the broadcasting, with the system controller **19**, via a second zonal speaker **122**, audio output for the second passenger **14**B at block **234**, further includes broadcasting an audible alert that the vehicle is nearing the destination as the vehicle approaches the destination. For example, the system controller **19** may provide an audible alert, such as“Your friend is on your left in 200 yards”.

(106) A system of the present disclosure offers several advantages. These include providing a floating image that is perceived by the passengers at a centrally location position within the vehicle compartment. This provides a camp-fire like viewing atmosphere where the passengers can all view a common floating image, or each passenger can view a unique floating image. Further, a system in accordance with the present disclosure provides the ability to display annotations and information not embedded within the virtual image and to ensure such annotations and information are perceived by a passenger at a proper location relative to the virtual image and in a plane between the passenger and the floating image. The system also allows a passenger to interact with the virtual image via the touch screen passenger interface and uses parallax compensation to ensure the system correctly correlates passenger input via the passenger interface to annotations and information displayed along with the virtual image.

(107) The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A method of using a system for generating a centrally located floating three-dimensional image display for a plurality of passengers positioned within a vehicle compartment, wherein the system is adapted to support interactive gaming between the first passenger and a remote opponent, comprising: displaying, with a first display of an image chamber in communication with a system controller, a first image; including, within the first image, an image of the remote opponent; receiving, with a first reflector individually associated with a first passenger, the first image from the first display; reflecting, with the first reflector, the first image to the first passenger, wherein the first passenger perceives the first image floating at a central location within the image chamber; displaying, with a second display of the image chamber in communication with a system controller, a second image; receiving, with a second reflector individually associated with a second passenger,

the second image from the second display; reflecting, with the second reflector, the second image to the second passenger, wherein the second passenger perceives the second image floating at the central location within the image chamber; displaying, with a transparent display in communication with the system controller and positioned between eyes of the first passenger and the first reflector and between the eyes of the second passenger and the second reflector, first private information to the first passenger within an image plane positioned in front of the first image floating at the central location within the image chamber and second private information to the second passenger within an image plane positioned in front of the second image floating at the central location within the image chamber; including, within the first private information displayed to the first passenger on the transparent display, game information; receiving, with the system controller, input from the first passenger and the second passenger; collecting, with an external scene camera, images of an external environment outside the vehicle compartment; displaying, with an augmented reality display in communication with the system controller and positioned within the vehicle compartment remotely from the image chamber, general information related to at least one of the first image, the second image, the first and second private information displayed on the transparent display and the input received from the first passenger and the second passenger; including, within the general information displayed on the augmented reality display, gameplay action.

2. The method of claim 1, wherein the augmented reality display includes a transparent substrate, having light emitting particles dispersed therein, positioned on a window within the vehicle compartment, the displaying, with the augmented reality display positioned within the vehicle compartment remotely from the image chamber, general information related to the first image, the private information displayed on the transparent display and the input received from the first passenger including: generating, with a primary graphic projection device in communication with the system controller, a first set of images upon the window within the vehicle compartment based on visible light, wherein the first set of images are displayed upon a primary area of the window; and generating, with a secondary graphic projection device in communication with the system controller, a second set of images upon a secondary area of the window based on an excitation light, wherein the light emitting particles in the windscreen emit visible light in response to absorbing the excitation light, and wherein the first set of images displayed upon the primary area of the windscreen cooperate with the second set of images displayed upon the secondary area of the windscreen to create an edge-to-edge augmented reality image.

3. The method of claim 2, wherein, the receiving, with the system controller, input from the first passenger and the second passenger, further includes: receiving, with the system controller, via the transparent display, input from the first passenger and the second passenger; receiving, with the system controller, via at least one first sensor, input comprising a position of a head and eyes of the first passenger; receiving, with the system controller, via at least one first gesture sensor, information related to gestures made by the first passenger; collecting, with the system controller, via a first microphone, audio input from the first passenger; and collecting, with the system controller, via a second microphone, audio input from the second passenger; and the method further including: broadcasting, with the system controller, via a first zonal speaker, audio output for the first passenger; and broadcasting, with the system controller, via a second zonal speaker, audio output for the second passenger.

4. The method of claim 1, further including: capturing, with an internal scene camera mounted within the vehicle compartment and in communication with the system controller, images of the augmented reality display; and sending, with the system controller, images of the augmented reality display to the remote opponent.

5. The method of claim 4, further including: receiving, with the system controller, via the at least one first sensor, input comprising a position of a head and eyes of the first passenger; receiving, with the system controller, via the at least one first gesture sensor, information related to gestures made by the first passenger; and sending, with the system controller, input from the first passenger

collected by the transparent display, the at least one first sensor and the at least one gesture sensor to a game system for incorporation into gameplay action.
6. The method of claim 5, further including supporting communication between the first passenger and the remote opponent with the first microphone and the first zonal speaker.
7. The method of claim 3, wherein the system is adapted to support a personalized driving tour for the plurality of passengers positioned within the vehicle compartment, the method including: collecting, with the system controller, information related to a first pre-determined site and a second pre-determined site located along a pre-determined path of a driving tour; collecting, with the system controller, information related to the first passenger and the second passenger; identifying, with the system controller, at least one first point of interest located between the first pre-determined site and the second pre-determined site and relevant to the first passenger and collecting information related to the at least one first point of interest; identifying, with the system controller, at least one second point of interest located between the first pre-determined site and the second pre-determined site and relevant to the second passenger and collecting information related to the at least one second point of interest; and when stopped at the first pre-determined site and when traveling between the first pre-determined site and the second pre-determined site, and when stopped at the second pre-determined site, including, with the system controller, the information related to the at least one first point of interest within the first image and including, with the system controller, the information related to the at least one second point of interest within the second image.
8. The method of claim 7, further including, when stopped at the first pre-determined site, including, with the system controller, information related to the first pre-determined site within the general information displayed by the augmented reality display.
9. The method of claim 3, further including, with the system controller: receiving, via the first microphone, a command from the first passenger; receiving, via the at least one first sensor and the at least one first gesture sensor data related to gestures made by the first passenger and the direction of a gaze of the first passenger; identifying, based on the data related to gestures made by the first passenger, the direction of the gaze of the first passenger and images of the external environment outside the vehicle compartment, an object outside of the vehicle compartment that the first passenger is looking at through the window with the augmented reality display; highlighting the object within the augmented reality display; and supporting, with the system controller, via the transparent display, interaction, by the first passenger, with the first image.
10. The method of claim 9, wherein, when the system controller determines that the object is a permanent object, the method includes: collecting information, with the system controller, from remote data sources, related to the object; and including, within the first private information, textual information related to the object.
11. The method of claim 9, wherein, when the system controller determines that the object is a non-permanent object, the method includes: identifying the object with object identification algorithms within the system controller; including, within the first private information, textual information related to the object.
12. The method of claim 3, further including: collecting, with the system controller, from remote data sources, information related to shopping interests of the first passenger; identifying an item that the first passenger may be interested in purchasing based on the information related to shopping interest of the first passenger; identifying, with the system controller, retailers that sell the item and are located one of, within a pre-determined distance, and located on a pre-determined route; collecting, with the system controller, from identified retailers, quantity and pricing information about the item; and including, within the first private information, the information related to retailers that sell the item, quantity and pricing if the item.
13. The method of claim 12, further including: receiving, with the system controller, via the transparent display, input from the first passenger expressing interest in the item; including, within

the first image, an image of the item; and highlighting, within the augmented reality display, an identified retailer, when an identified retailer is visible through the window and the augmented reality display.

14. The method of claim 3, further including: receiving, via the transparent display, input from the first passenger including augmentations to the first image; receiving, with the system controller, from the first passenger, input indicating that the first passenger is finished and wants to display the first image and the augmentations within the augmented reality display; and including, within the general information displayed on the augmented reality display, the first image and augmentations by the first passenger.

15. The method of claim 3, further including: receiving, from the first passenger, input on a destination; including, in the second private information displayed on the transparent display for the second passenger, information about the destination; and highlighting, within the augmented reality display, the destination when the destination is visible through the window and the augmented reality display.

16. The method of claim 1, further including: receiving, from the first passenger, input on a destination; including, in the second private information displayed on the transparent display for the second passenger, information about the destination; and highlighting, within the augmented reality display, the destination when the destination is visible through the window and the augmented reality display.

17. A system for generating a centrally located floating three-dimensional image display for a plurality of passengers positioned within a vehicle compartment within a vehicle, comprising: a system controller; an image chamber including: a first display adapted to project a first image; a first reflector individually associated with the first display and a first one of the plurality of passengers, the first reflector adapted to receive the first image from the first display and to reflect the first image to the first passenger, wherein the first passenger perceives the first image floating at a central location within the image chamber; a second display adapted to project a second image; and a second reflector individually associated with the second display and a second one of the plurality of passengers, the second reflector adapted to receive the second image from the second display and to reflect the second image to the second passenger, wherein, the second passenger perceives the second image floating at the central location within the image chamber; and a transparent touch screen display positioned between the first reflector and the first passenger and between the second reflector and the second passenger and adapted to display first private information to the first passenger within an image plane positioned in front of the first image floating at the central location within the image chamber and to receive input from the first passenger, and adapted to display second private information to the second passenger within an image plane positioned in front of the second image floating at the central location within the image chamber and to receive input from the second passenger; an external scene camera adapted to collect images of an external environment outside the vehicle compartment; an augmented reality display positioned within the vehicle compartment remotely from the image chamber and adapted to display general information related to at least one of the first image, first and second private information displayed on the transparent display and input received from the first and second passengers; and an internal scene camera mounted within the vehicle compartment and in communication with the system controller and adapted to capture images of the augmented reality display.

18. The system of claim 17, wherein the augmented reality display includes: a transparent substrate, having light emitting particles dispersed therein, positioned on a window within the vehicle compartment; a primary graphic projection device for generating a first set of images upon the window of the vehicle based on visible light, wherein the first set of images are displayed upon a primary area of the window; a secondary graphic projection device for generating a second set of images upon a secondary area the window of the vehicle based on an excitation light, wherein the

light emitting particles in the window emit visible light in response to absorbing the excitation light, and wherein the first set of images displayed upon the primary area of the window cooperate with the second set of images displayed upon the secondary area of the window to create an edge-to-edge augmented reality view of a surrounding environment of the vehicle; a primary graphics processing unit in electronic communication with the primary graphic projection device and the system controller; and a secondary graphics processing unit in electronic communication with the secondary graphic projection device and the system controller.

19. The system of claim 18, wherein the system is selectively moveable vertically up and down along a vertical central axis, the first display and the first reflector are unitarily and selectively rotatable about the vertical central axis, and the second display and the second reflector are unitarily and selectively rotatable about the vertical central axis; the system further including: first sensors adapted to monitor a position of a head and eyes of the first passenger, wherein, the first display and first reflector are adapted to rotate in response to movement of the head and eyes of the first passenger, and second sensors adapted to monitor a position of a head and eyes of the second passenger, wherein, the second display and the second reflector are adapted to rotate in response to movement of the head and eyes of the second passenger, the system adapted to move up and down along the vertical axis in response to movement of the head and eyes of the first passenger and movement of the head and eyes of the second passenger; and a first gesture sensor adapted to gather information related to gestures made by the first passenger, and a second gesture sensor adapted to gather information related to gestures made by the second passenger, wherein, the system is adapted to receive input from the first and second passengers via data collected by the first and second gesture sensors.

20. A method of using a system for generating a centrally located floating three-dimensional image display for a plurality of passengers positioned within a vehicle compartment within a vehicle, comprising: displaying, with a first display of an image chamber in communication with a system controller, a first image; receiving, with a first reflector individually associated with a first passenger, the first image from the first display; reflecting, with the first reflector, the first image to the first passenger, wherein the first passenger perceives the first image floating at a central location within the image chamber; displaying, with a second display of the image chamber in communication with a system controller, a second image; receiving, with a second reflector individually associated with a second passenger, the second image from the second display; reflecting, with the second reflector, the second image to the second passenger, wherein the second passenger perceives the second image floating at the central location within the image chamber; displaying, with a transparent display in communication with the system controller and positioned between eyes of the first passenger and the first reflector and between the eyes of the second passenger and the second reflector, first private information to the first passenger within an image plane positioned in front of the first image floating at the central location within the image chamber and second private information to the second passenger within an image plane positioned in front of the second image floating at the central location within the image chamber; receiving, with the system controller, via the transparent display, input from the first passenger and the second passenger; receiving, with the system controller, via at least one first sensor, input comprising a position of a head and eyes of the first passenger; receiving, with the system controller, via at least one first gesture sensor, information related to gestures made by the first passenger; collecting, with the system controller, via a first microphone, audio input from the first passenger; broadcasting, with the system controller, via a first zonal speaker, audio output for the first passenger; collecting, with the system controller, via a second microphone, audio input from the second passenger; and broadcasting, with the system controller, via a second zonal speaker, audio output for the second passenger collecting, with an external scene camera, images of an external environment outside the vehicle compartment; collecting, with an internal scene camera, images of the augmented reality display; and displaying, with an augmented reality display in communication with the system

controller and having a transparent substrate, having light emitting particles dispersed therein, positioned on a window within the vehicle compartment, general information related to at least one of the first image, the second image, the first and second private information displayed on the transparent display and the input received from the first passenger and the second passenger by: generating, with a primary graphic projection device in communication with the system controller, a first set of images upon the window within the vehicle compartment based on visible light, wherein the first set of images are displayed upon a primary area of the window; and generating, with a secondary graphic projection device in communication with the system controller, a second set of images upon a secondary area of the window based on an excitation light, wherein the light emitting particles in the windscreen emit visible light in response to absorbing the excitation light, and wherein the first set of images displayed upon the primary area of the windscreen cooperate with the second set of images displayed upon the secondary area of the windscreen to create an edge-to-edge augmented reality image.
